Mantis IP Visualization: Patent Dataset Overview

HUPD and Lens pipelines, datasets, and schemas

Mantis IP Visualization

May 2026

Contents

1 Introduction

This document covers the two patent datasets we loaded for Mantis: the Harvard USPTO Patent Dataset (HUPD) and a targeted export from Lens.org. For each one we describe what the source is, how we pulled and cleaned a usable subset, and what the resulting files look like. The Lens section also covers how we produced `US-lens-clean.json` (US-only records, cleaned and flattened) and `lens-db-translated.json` (the broader multilingual export translated to English).

Part I

Harvard USPTO Patent Dataset (HUPD)

2 What is HUPD?

HUPD is a collection of 4.5 million US patent applications filed between 2004 and 2018, stored with the original inventor-submitted text rather than the final granted version. It includes both accepted and rejected applications across all technology domains, which makes it well-suited for patentability work.

3 What's in the full dataset

Each record is a JSON file with 26 fields:

Field	What it is
application_number	USPTO application identifier
publication_number	Publication number once the app is published
title	Short title of the invention
decision	ACCEPTED, REJECTED, or PENDING
date_produced	Date the record was produced internally
date_published	Date the application was published
filing_date	Date filed with the USPTO
patent_issue_date	Date granted (if accepted)
abandon_date	Date abandoned (if applicable)
patent_number	Granted patent number (if accepted)
main_cpc_label	Primary CPC code
cpc_labels	All CPC codes
main_ipcr_label	Primary IPC code
ipcr_labels	All IPC codes
uspc_class	US Patent Classification class
uspc_subclass	US Patent Classification subclass
examiner_id	USPTO examiner ID
examiner_name_last	Examiner last name
examiner_name_first	Examiner first name
examiner_name_middle	Examiner middle name
inventor_list	Inventors (name, city, state, country)
abstract	Abstract as written by the applicant
claims	Claims section
background	Background of the invention
summary	Summary of the invention
full_description	Full description as filed

The data is split into per-year `.tar.gz` archives, roughly 62 GB uncompressed in total.

4 How we got our 2k sample

4.1 Sampling approach

The Hugging Face `datasets` loader no longer works with recent library versions, so we pulled records directly from the per-year archives over HTTP via `data-load.py`, reading them as streams to avoid downloading the full 62 GB. We took roughly 140 records from the front of each year's archive (2004–2018), shuffled, and trimmed to 2,100. Archive order within a year is unsorted by application number, so this is a reasonable stand-in for a random draw.

4.2 Data cleaning

Records with fewer than 25 words in the abstract were dropped—anything shorter is a stub or formatting artifact. The 2,100 records here were already pre-filtered, so all 2,100 remain.

4.3 Final dataset at a glance

Property	Value
Total records	2,100
Filing years covered	2004–2018 (15 years)
Records per year	∼140 (roughly even)
Fields per record	26
Unique examiners	1,777
Mean inventors/patent	2.74
Mean IPC labels/patent	2.08
Abstract: median length	110 words
Claims: median length	821 words
Full description: median	5,998 words
Total words (full desc.)	∼17.8 million

Table 1: Summary statistics for the HUPD 2k sample.

`title`, `abstract`, `claims`, and `full_description` are all present in 100% of records. `background` is present in 94.6% and `summary` in 93.3%—both can be legitimately absent on older or shorter applications.

5 The data in detail

5.1 IPC and CPC classification codes

IPC (International Patent Classification) is the older WIPO system, present on 93.7% of records. Codes are hierarchical: a letter for the top-level section, a two-digit class, a subclass letter, and further subdivisions. `G06F`, for example, is G (Physics) → 06 (Computing) → F (Electrical digital data processing).

CPC (Cooperative Patent Classification) was introduced jointly by the USPTO and EPO around 2013, so it only covers 40.2% of the sample—roughly the filings from 2013 onward. It follows the same letter-section structure as IPC.

The eight top-level WIPO sections:

Section	What it covers
A	Human Necessities
B	Operations and Transport
C	Chemistry and Metallurgy
D	Textiles and Paper
E	Fixed Constructions
F	Mechanical Engineering
G	Physics / Computing
H	Electricity / Electronics

The top 20 IPC subclasses by primary label count:

Subclass	Description	Count
G06F	Electrical digital data processing	176
H01L	Semiconductor devices	117
A61K	Medical / dental / personal care preparations	76
H04L	Transmission of digital information	70
H04N	Pictorial communication	61
A61B	Diagnosis / surgery / medical instruments	51
H04W	Wireless communications networks	42
G01N	Investigating materials	39
G06Q	Data processing for administrative / commercial purposes	37
A61F	Implants / prosthetics	28
G02B	Optical elements	27
G06K	Graphical data reading / recognition	27
H04B	Transmission (general)	24
B41J	Selective printing mechanisms	23
A61M	Devices for introducing media into the body	19
G11C	Static stores (memory)	19
H05K	Printed circuits / manufacture of assemblies	19
G03G	Electrography / electrophotography	18
H01R	Electrically-conductive connections	17
F16H	Gearing	16

5.2 Category and outcome distributions

The sample skews toward G (Physics) and H (Electricity), consistent with the general trend in US filings over this period. All eight sections are represented. Computing (G06F) is the largest subclass, followed by semiconductors (H01L) and pharmaceuticals (A61K). There are 319 distinct subclasses in total.

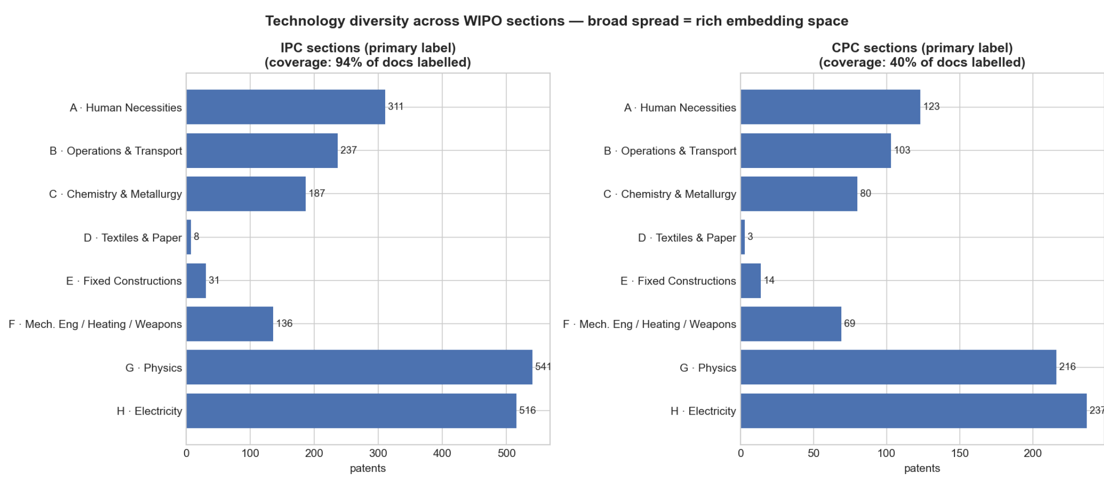

Figure 1: Distribution across the eight WIPO top-level sections for IPC (94% coverage) and CPC (40% coverage).

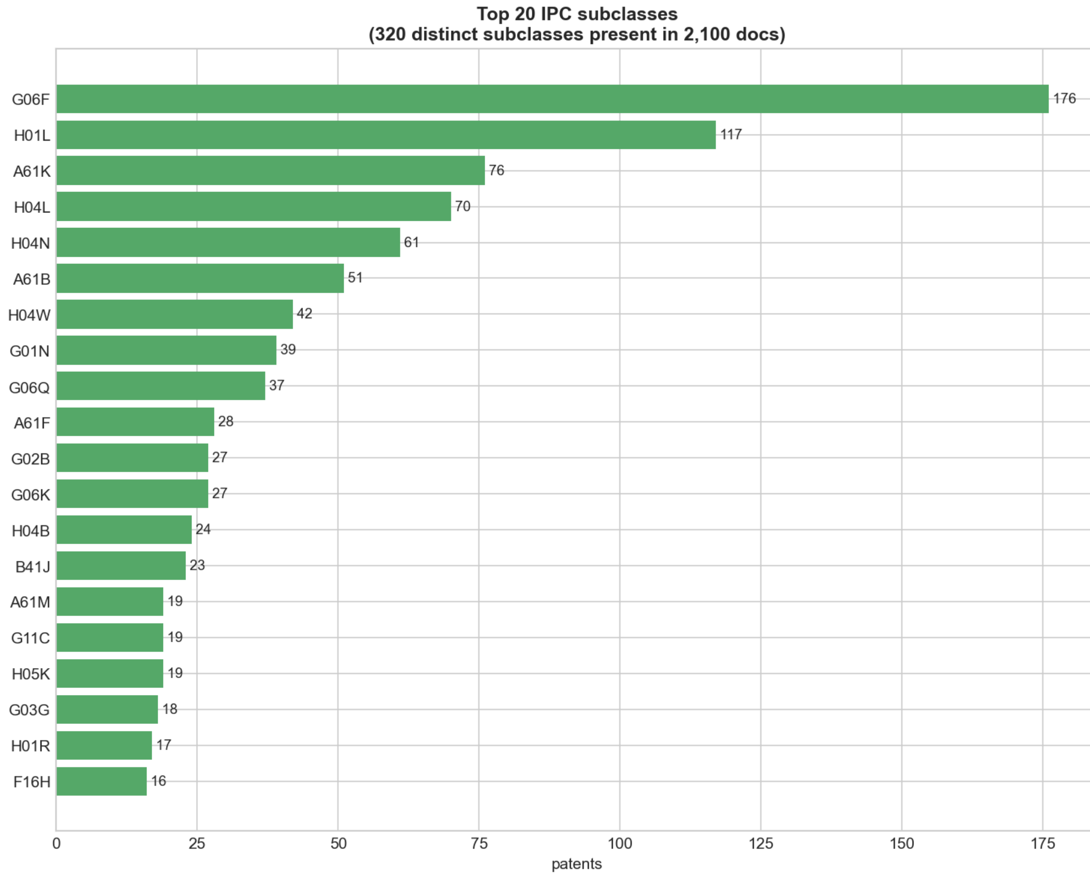

Figure 2: Top 20 IPC subclasses (319 distinct subclasses present overall).

Just over half the sample (54.6%) was accepted, about a quarter (25.9%) rejected, and 19.6% are still pending. The pending share rises sharply from 2016 onward—those applications had not resolved when the dataset was frozen in 2022.

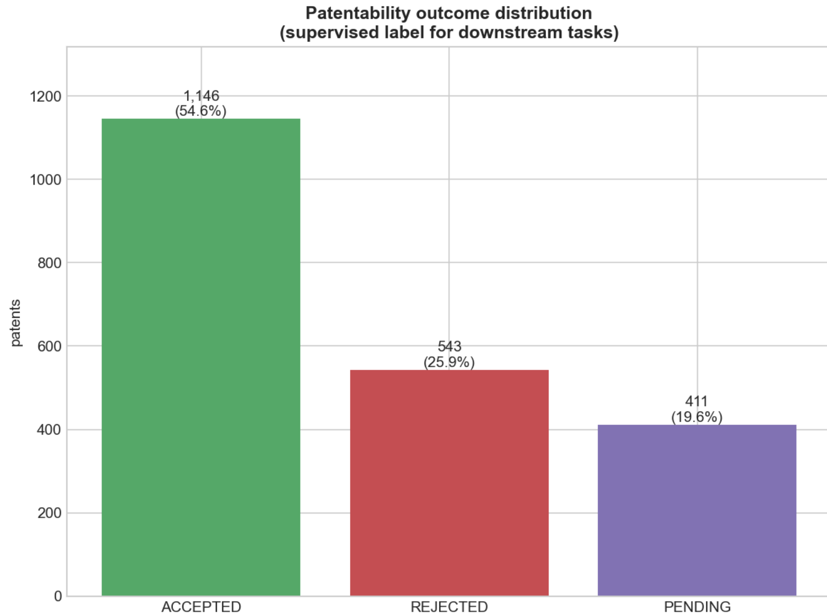

Figure 3: Patentability outcomes across the 2,100-record sample.

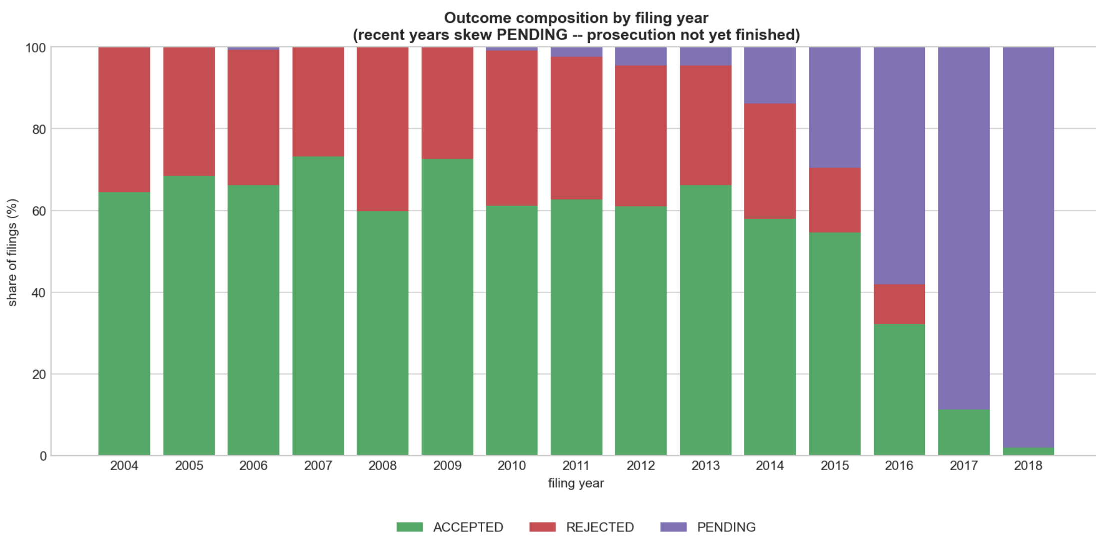

Figure 4: Outcome composition by filing year. Recent years skew PENDING because prosecution had not finished when the dataset was frozen.

5.3 Text length details

Field	Non-empty	Mean	Median	Std	p90	Max
title	100%	8	7	5	14	44
abstract	100%	111	110	43	151	405
claims	100%	964	821	723	1,655	11,589
background	95%	483	345	584	1,040	9,290
summary	93%	742	415	1,277	1,624	22,730
full_description	100%	8,466	5,998	8,871	15,435	131,338

Table 2: Word-count statistics per text field. There are notable outliers throughout.

Part II

Lens Patent Database

6 What is the Lens Dataset?

Lens.org is a free, open-access patent search platform run by Cambia. It pulls records from the USPTO, EPO, WIPO, JPO, CNIPA, and more—over 150 million patents total. Each record has bibliographic metadata, legal status, family linkages, classification codes, and (where available) abstract, description, and claims. The Lens API supports bulk JSON export, which is how we got our data.

7 What's in the raw dataset

The raw export (`lens-raw.jsonl`) has **8,357 records** from a targeted query on IP relevant to Mantis. Records span 43 jurisdictions and 20 languages, running from the late 19th century through 2024, though the bulk are from 2020 onward.

7.1 Jurisdictions and languages

Non-US records make up the majority of the pull, with China, Japan, and the EPO being the largest non-US sources (see Figure 5). The multilingual character is a direct result of the broad query scope and gets resolved during cleaning.

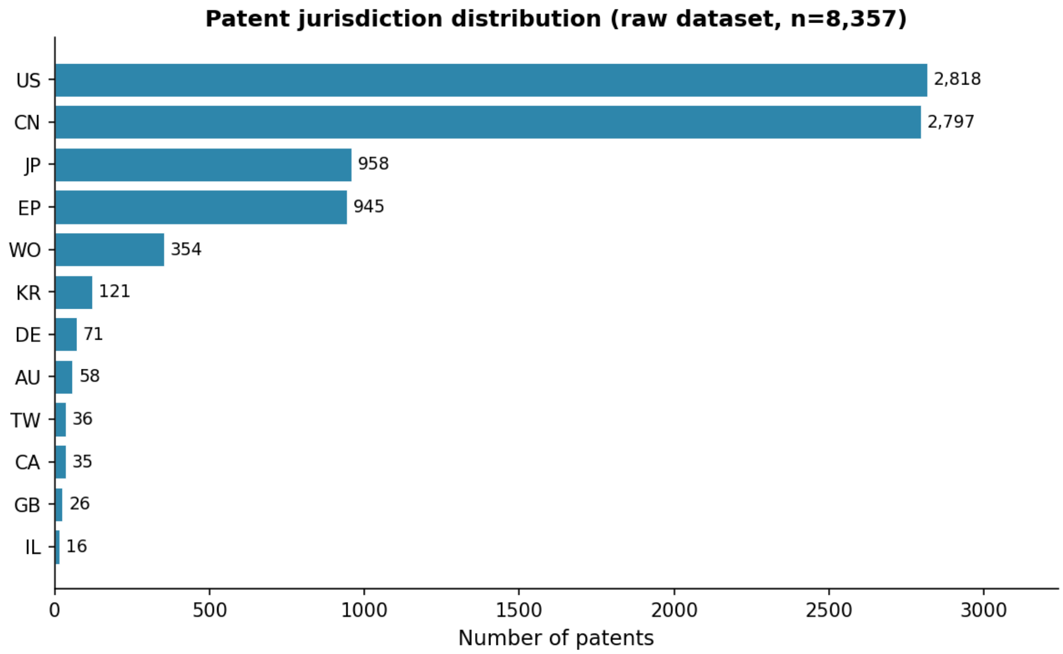

Figure 5: Jurisdiction distribution (top 12 offices, n = 8,357).

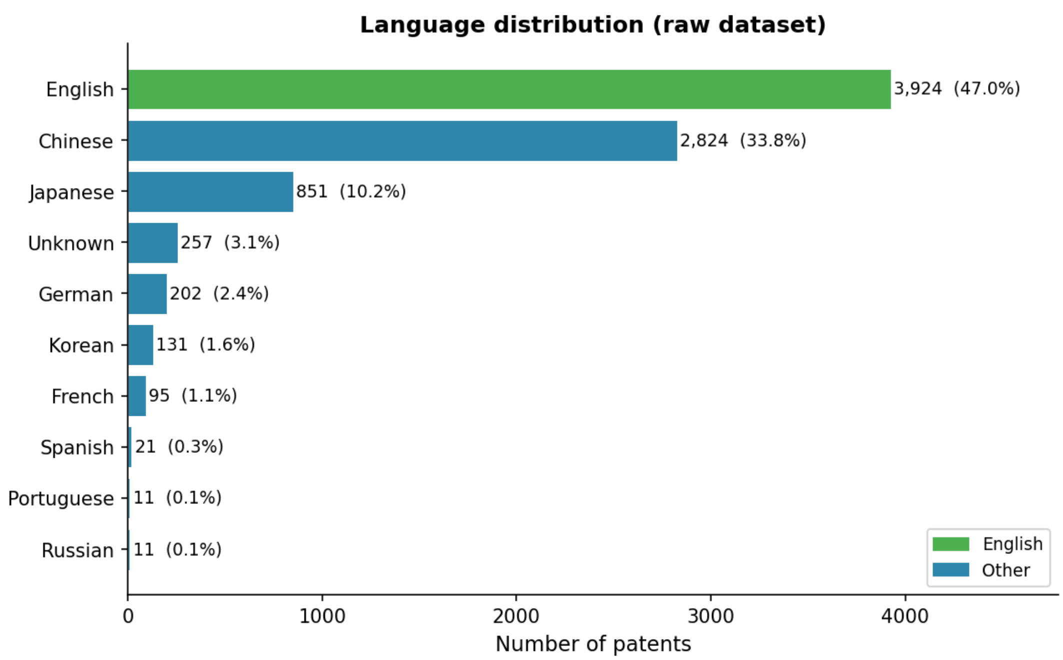

Figure 6: Language distribution across the raw dataset.

7.2 Field completeness

The abstract is the most reliably present field. Description and claims are missing from over half of records—normal for non-US or older filings. The sequence listing field has almost no coverage and was dropped.

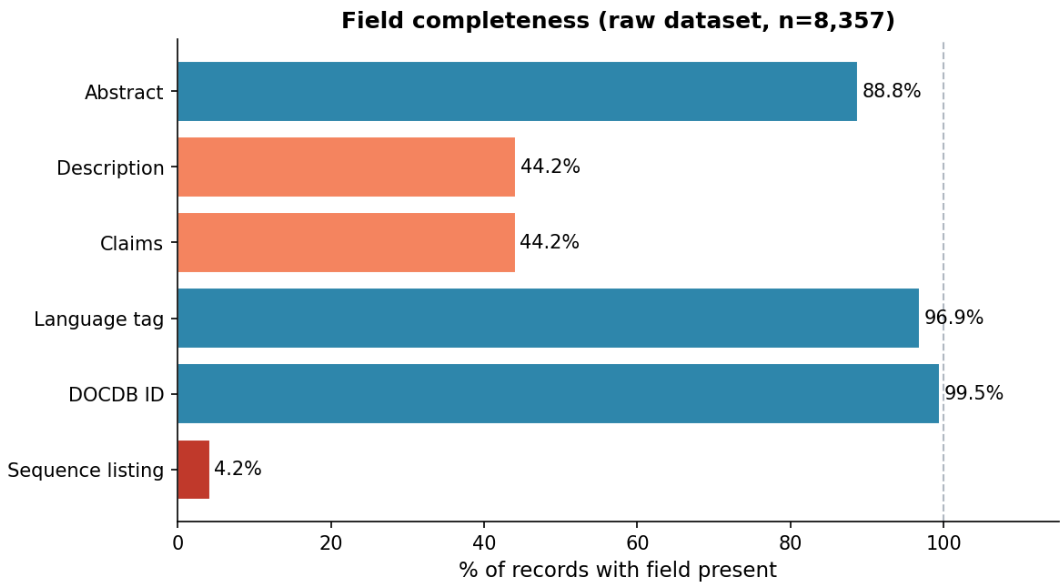

Figure 7: Percentage of raw records with each field populated.

7.3 Publication types and legal status

The dataset mixes applications and granted patents. Most are either active or pending, reflecting the recency of the pull.

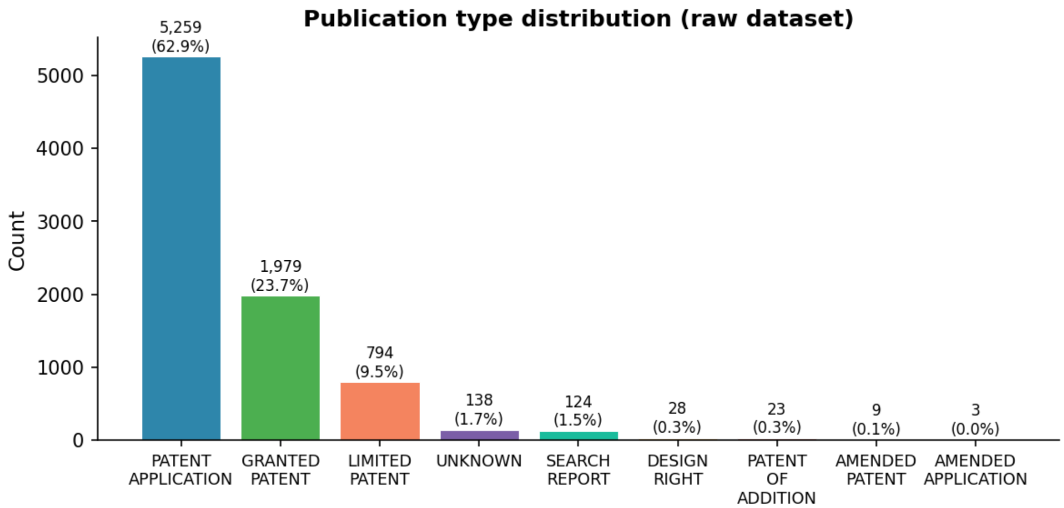

Figure 8: Publication type distribution.

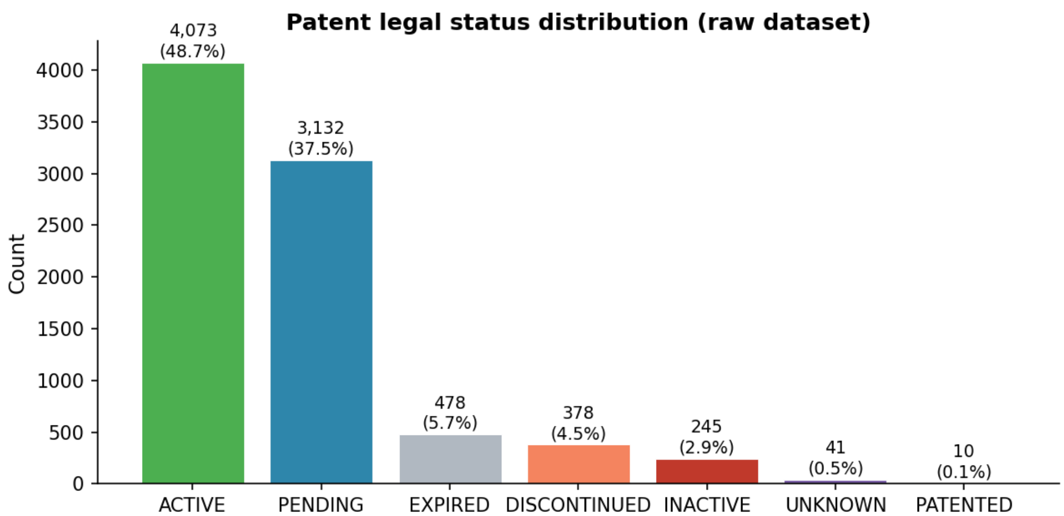

Figure 9: Legal status distribution.

7.4 Publication year

The dataset is heavily weighted toward recent filings, with most records from 2020 onward and a thin historical tail going back to the 19th century.

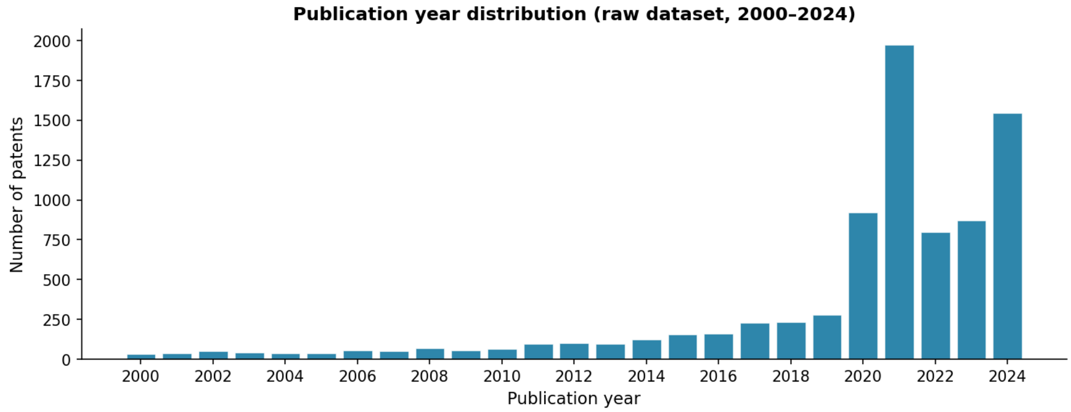

Figure 10: Publication year distribution (2000–2024).

7.5 Technology coverage (IPC sections)

All eight WIPO sections are present. G (Physics / Computing) and H (Electricity) have the highest counts, consistent with the Mantis technology focus.

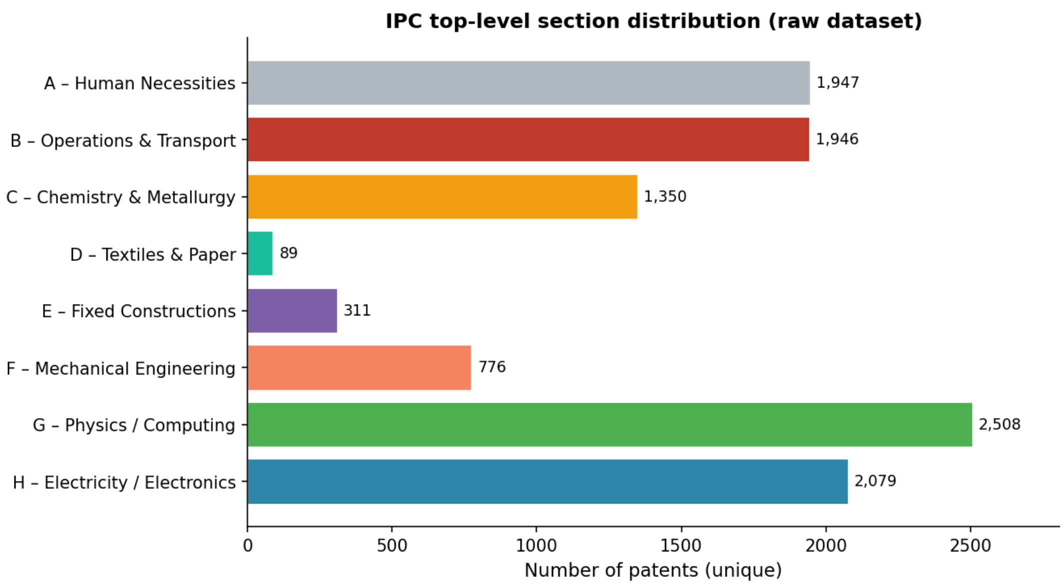

Figure 11: Unique patents per IPC top-level section (a patent may appear in multiple).

8 Producing `US-lens-clean.json`

Starting from `lens-raw.jsonl`, we applied three sequential filters and then flattened the nested JSON schema to produce `US-lens-clean.json`.

8.1 Filtering

1. **US jurisdiction only.** Dropping non-US records cut the dataset from 8,357 to 2,818 and, as a side effect, resolved the language issue: 99.9% of what remained was in English (three records had no language tag but are presumed English given their US jurisdiction).
2. **Abstract required.** The abstract is Mantis's primary semantic signal, so records without one were dropped. Only 39 of the 2,818 US records lacked an abstract, leaving 2,779.
3. **Deduplication.** Records were deduplicated on `lens_id`. No duplicates were found, so the count stayed at 2,779.

Stage	**Records**
Raw `lens-raw.jsonl`	8,357
US jurisdiction only	2,818
Abstract present	2,779
Deduplicated	2,779

8.2 Schema flattening

Lens records are deeply nested—inventor names, for instance, are buried at `biblio > parties > inventors > extracted_name > value`. `clean.py` flattens everything to a single-level schema of 19 fields, dropping any fields that are redundant or not relevant for patent identification or semantic analysis. Missing values are filled with the string `"Null"` so every record has the same shape.

9 Final dataset at a glance

Property	**Value**
Total records	2,779
Jurisdiction	US only
Language	English (99.9%), unknown (0.1%)
Fields per record	19
Records with abstract	2,779 (100%)
Records with description	1,196 (43.0%)
Abstract: mean length	105 words
Abstract: median length	107 words
Abstract: 90th pctile	149 words

Table 3: Summary statistics for `US-lens-clean.json`.

10 Schema reference

Field	Description
lens_id	Globally unique Lens.org record identifier.
jurisdiction	Patent office (always US here).
doc_number	USPTO document number.
kind	Publication stage code (e.g., A, B2).
date_published	Publication date (YYYY-MM-DD).
lang	Language of the document.
publication_type	e.g., PATENT_APPLICATION, GRANTED_PATENT.
patent_status	e.g., ACTIVE, PENDING, EXPIRED.
title	Invention title(s); may include multiple translations.
application_date	Date filed with the USPTO.
priority_date	Earliest priority claim date.
applicants	Applicant / assignee names.
inventors	Inventor names.
ipc_classifications	IPC classification symbols.
cited_by_lens_ids	Forward citations (Lens IDs).
simple_family_lens_ids	Simple patent family members (Lens IDs).
extended_family_lens_ids	Extended patent family members (Lens IDs).
abstract	Full abstract text.
description	Full description text, where available.

Table 4: The 19 fields in `US-lens-clean.json`. Missing values are stored as `"Null"`.

11 Producing `lens-db-translated.json`

`US-lens-clean.json` only keeps US records, which are almost entirely English. The broader cleaned export still has records in Chinese, Japanese, German, Korean, and others. To make that dataset usable by English-language models, we produced `lens-db-translated.json` by running a translation pass over `lens-db-clean.json`.

Translation used **Argos Translate**, a free offline neural machine translation library. Because translating thousands of records takes significant time, the job was left running on a VM overnight. For `title` and `abstract`—which are stored as lists of per-language entries in the raw Lens format—the pipeline picks any existing English entry first and only calls the translator when none is present. The `description` field is translated only for records whose language tag is not already English. After the pass, all text fields are plain strings and every record's `lang` is set to `"en"`.